

Machine Learning Classification Lab

Phishing Website Detection

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1 Introduction

This project uses the **Phishing Website Detection Dataset** from Kaggle containing 11,430 websites with 87 numeric features extracted from URLs and web content. The dataset is perfectly balanced with 5,715 legitimate and 5,715 phishing samples, has no missing values or duplicates, making it ideal for classification tasks.

Objective: Build and evaluate machine learning models to automatically detect phishing websites. This is critical for cybersecurity as phishing attacks steal sensitive user information. We trained 8 different classifiers and compared their performance using accuracy, precision, recall, F1-score, and confusion matrices.

2 Preprocessing Pipeline

Data Cleaning: Verified no missing values (0) and no duplicates (0). Removed the raw URL text column as it's unsuitable for numerical algorithms.

Encoding: Target variable encoded as: **legitimate** $\rightarrow$ 0, **phishing** $\rightarrow$ 1. All 87 features were already numeric.

Scaling: Applied StandardScaler ($z = \frac{x-\mu}{\sigma}$) to normalize features, essential for distance-based algorithms (KNN, SVM).

Splitting: 80% training (9,144 samples) and 20% testing (2,286 samples) with stratification to maintain class balance.

3 Models Used

Eight classification algorithms were trained:

1. **k-Nearest Neighbors (k=5):** Classifies based on majority vote of 5 nearest neighbors
2. **Naïve Bayes (Gaussian):** Probabilistic classifier using Bayes' theorem with independence assumption
3. **Decision Tree (max_depth=10):** Tree-based model splitting data by information gain
4. **SVM (RBF kernel):** Finds optimal hyperplane maximizing margin between classes
5. **Random Forest (100 trees):** Ensemble of decision trees using bagging
6. **Logistic Regression:** Linear classifier using sigmoid function
7. **Gradient Boosting (100 estimators):** Sequential ensemble correcting previous errors
8. **AdaBoost (100 estimators):** Adaptive boosting focusing on misclassified samples

4 Results

4.1 Cross-Validation (5-Fold)

Table 1: Cross-Validation Results on Training Set

Model	Mean Accuracy	Std Dev
Random Forest	96.47%	0.0020
Gradient Boosting	96.33%	0.0013
SVM	95.65%	0.0018
AdaBoost	95.06%	0.0055
Logistic Regression	94.62%	0.0035
Decision Tree	93.88%	0.0038
KNN	93.49%	0.0051
Naive Bayes	68.35%	0.0059

4.2 Test Set Performance

Table 2: Test Set Evaluation Metrics

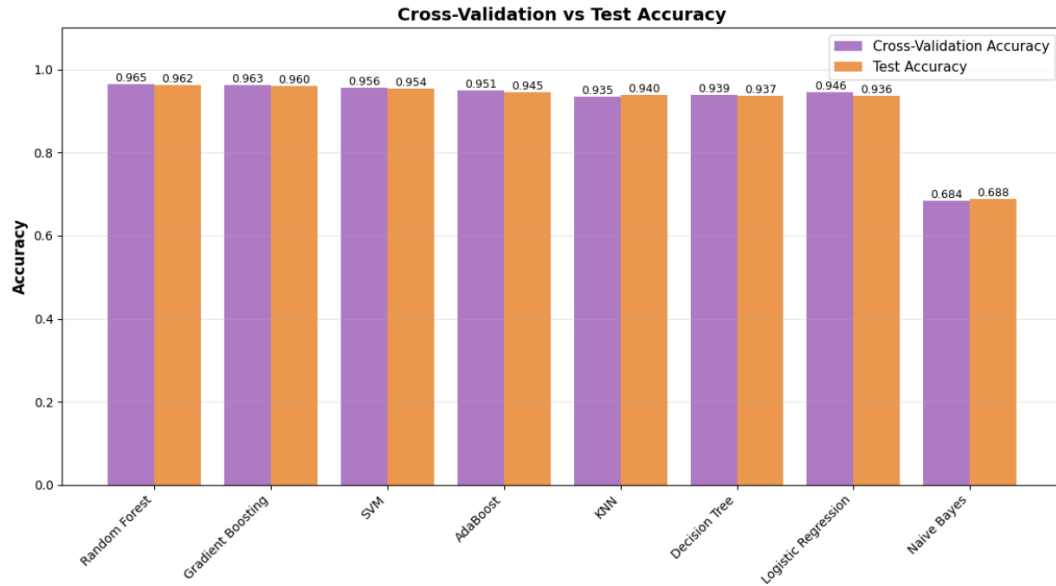
Model	Accuracy	Precision	Recall	F1
Random Forest	96.19%	95.83%	96.59%	96.21%
Gradient Boosting	96.02%	95.74%	96.33%	96.03%
SVM	95.41%	95.77%	95.01%	95.39%
AdaBoost	94.49%	94.80%	94.14%	94.47%
KNN	93.96%	95.39%	92.39%	93.87%
Decision Tree	93.70%	93.93%	93.44%	93.68%
Logistic Regression	93.61%	93.92%	93.26%	93.59%
Naive Bayes	68.81%	94.24%	40.07%	56.23%

4.3 Confusion Matrices

Table 3: Confusion Matrix Results (TN=True Neg, FP=False Pos, FN=False Neg, TP=True Pos)

Model	TN	FP	FN	TP
Random Forest	1095	48	39	1104
Gradient Boosting	1094	49	42	1101
SVM	1095	48	57	1086
AdaBoost	1084	59	67	1076
KNN	1092	51	87	1056
Decision Tree	1074	69	75	1068
Logistic Regression	1074	69	77	1066
Naive Bayes	1115	28	685	458

Random Forest Error Analysis: False Positive Rate = 4.2% (48/1143), False Negative Rate = 3.4% (39/1143), Total Errors = 3.8% (87/2286).



5 Discussion & Conclusion

5.1 Model Selection: Random Forest

Random Forest is selected as the best model based on comprehensive evaluation.

Justification:

1. **Highest Accuracy (96.19%):** Best performance among all 8 models with only 87 errors out of 2,286 test samples
2. **Best F1-Score (96.21%):** Optimal balance between precision and recall
3. **Superior Recall (96.59%):** Critical for security - detects 96.59% of phishing sites, missing only 39 attacks
4. **Excellent Generalization:** Small gap between CV (96.47%) and test accuracy (96.19%) shows no overfitting
5. **Lowest False Negatives (39):** Minimizes the most dangerous error type in cybersecurity
6. **Acceptable False Positives (48):** Only 4.2% of legitimate sites flagged, maintaining good user experience

5.2 Key Findings

Ensemble Methods Dominate: Top 3 models (Random Forest, Gradient Boosting, AdaBoost) are all ensembles, achieving >94% accuracy. This confirms that combining multiple learners significantly improves performance.

Naive Bayes Fails: Only 68.81% accuracy with catastrophic 40.07% recall. The independence assumption is violated, making it unsuitable for this dataset where features are highly correlated.

Feature Scaling Impact: SVM and KNN benefited significantly from StandardScaler, achieving 95.41% and 93.96% accuracy respectively.

Random Forest vs Others:

- vs Gradient Boosting: +0.17% accuracy, better recall (96.59% vs 96.33%)

- vs SVM: +0.78% accuracy, significantly better recall (96.59% vs 95.01%)
- vs AdaBoost: +1.70% accuracy, more stable (lower CV variance)

5.3 Practical Implications

The Random Forest model achieves deployment-ready performance:

- **Security:** 96.59% recall protects users from most phishing threats
- **Usability:** 95.83% precision minimizes false alarms
- **Reliability:** Consistent cross-validation scores demonstrate stability

5.4 Limitations & Future Work

Limitations: Dataset represents static features and may not capture emerging phishing techniques; computational cost of 100-tree ensemble.

Future Improvements: Implement deep learning (LSTM/CNN), add real-time dynamic features (SSL validity, DNS records), deploy with continuous retraining schedule.

5.5 Conclusion

This lab successfully demonstrated the complete ML workflow for phishing detection. Through systematic preprocessing, training of 8 algorithms, and comprehensive evaluation using cross-validation and multiple metrics, Random Forest emerged as the optimal classifier with 96.19% accuracy, 96.21% F1-score, and critically, 96.59% recall.

The high recall minimizes false negatives - the most dangerous error in cybersecurity - while maintaining excellent precision for user experience. The model's consistent performance across cross-validation folds and test set confirms its generalization capability, making it suitable for production deployment in cybersecurity applications.